

- (i) A coincident site lattice with six adsorbate species corresponding to eleven substrate units in the $[1\bar{1}0]$ direction.
- (ii) A badly ordered (1×2) phase.
- (iii) A disordered phase.
- e) The LEED sequence is symptomatic of little or no dipole-dipole interaction between the adsorbed entities.
- f) The authors favour a dissociative chemisorption mechanism but reconstruction effects and interaction with the bulk cannot, at this stage, be ruled out.

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Quantum-Mechanical Theory of Reaction Rates

By S. G. Christov

Institute of Physical Chemistry, Bulgarian Academy of Sciences, Sofia, Bulgaria

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A generalization of previous studies in chemical kinetics, including nuclear tunneling, for a system of n -dimensions is presented. On the basis of a quantum-mechanical formulation a general equation for the reaction rate is derived which is free of the restrictions of activated-complex theory. The conditions on which the reaction proceeds classically are determined by introducing a "characteristic temperature" for which, within certain assumptions, a simple expression is found. Several general consequences from the fundamental rate equation are discussed.

1. Introduction

One of the most useful formulations of the theory of reaction rates is developed in the framework of Eyring-Polanyi's activated-complex theory [1] which has been widely and successfully applied in chemical kinetics. This theory is based on the concept of a potential-energy surface with a "col" and a saddle-point determining the transition state of the system

in which it is considered as an "activated complex" like a relatively stable molecule. A particular kind of thermal equilibrium between initial and transition states (reactants and activated complexes) is thus postulated. The movement of nuclei is treated classically and the reaction is described by the translational motion of a mass-point along a line in the nuclear configuration space, passing through the saddle-

point. This line is called the "reaction coordinate" and is assumed to be a straight line at least in the saddle-point region.

Many efforts have been made to justify the basic assumptions of activated-complex theory. The calculations of the potential energy as a function of nuclear coordinates implies the validity of Ehrenfest adiabatic principle which permits to separate the motions of nuclei and electrons using the Born-Oppenheimer approximation. Earlier half-empirical computations [1, 2] have been recently replaced by more accurate *a priori* calculations [3]. They all reveal the same qualitative features of potential-energy surfaces which support the general idea of activated-complex theory that there exists a col on these surfaces (at least for triatomic collinear reactions of the type $H_2 + H \rightarrow H + H_2$). However, the "reaction path", i.e. the line of lowest energy, is shown to be non-rectilinear in the saddle-point region.

The equilibrium postulate has been usually justified [4, 5] by assuming that during reaction the system remains in the same quantum state in respect to all coordinates except the reaction coordinate. Thus, the frequency of a vibration normal to the reaction path changes adiabatically thereby assuring an equilibrium distribution of the vibrational energy in the transition state. This permits to use quantum-statistical methods to describe the vibrations of the activated complex. However, many years ago Kassel [6] has pointed out that the lifetime of the transition state is very short, hence it seems doubtful whether a stationary quantum-mechanical treatment of the activated complex is fully justified. This conclusion is supported by recent dynamical calculation, both classical [7] and quantum-mechanical [8], which show the reaction $H_2 + H \rightarrow H + H_2$ to be non-adiabatic with respect to the symmetric stretch vibration [8]. Thus, it appears that there is no full quantization of the vibrational states of the activated complex [5, 9].

The classical treatment of the motion along the reaction coordinate is certainly correct for adiabatic reactions at sufficiently high temperatures, however, at low temperatures a more accurate quantum-mechanical treatment will be necessary. This is especially the case of hydrogen atom (proton)-transfer processes already at ordinary temperatures [10–24].

Quantum-mechanical corrections to the classical reaction rate have been usually made by writing corresponding factors to the rate equation of transition state theory [1]. Thus, a "transmission coefficient" is simply added to account for the reflection due to the curvature of the reaction path or for the nonadiabatic transitions from the lower to the upper potential surface [1, 25]. Similarly, a "tunneling factor" is written as a correction for the deviation from the classical motion due to nuclear tunneling [1, 10].

The tunneling has been usually treated in one-dimensional approximation, i.e. by assuming a rectilinear reaction coordinate [10–19]. In general, this assumption is justified only in the vicinity of the saddle-point so that such treatment is sufficiently accurate at relatively high temperatures. Two-dimensional tunneling corrections for collinear reactions

$AX + B = A + XB$ have been recently computed using different methods [20–24].

Eyring et al. [25] have proposed a quantum-mechanical formulation of reaction rate theory in which the assumption of equilibrium between initial and transition state is removed. In this way, however, the usual form of the rate equation with missing tunneling correction is obtained. In a recent generalization of activated-complex theory Marcus [26, 27] has treated the classical and quantum mechanics of reaction rates in a near vibrationally adiabatic approximation, using curvilinear coordinates, which permit to include tunneling in a wider region around the saddle-point than do the cartesian ones.

In a series of previous papers [13–18] the role of nuclear tunneling in chemical and electrochemical reactions has been considered by assuming a rectilinear reaction coordinate. The specific concepts of activated-complex theory have not been used in these investigations; thus a statistical equilibrium only in the initial state of the system is assumed. A generalized WKB expression for the transition probability [12, 14, 18, 28, 29] has been employed and a "characteristic temperature" [14a, 16, 18] was introduced as a very convenient criterion for determining the conditions at which the tunneling is important or negligible. This criterion has been successfully used in various problems of nuclear tunneling in chemical kinetics [14–18] and electron tunneling in solid state physics [30, 31] and biophysics [32].

In the present paper a generalization of the previous results of chemical kinetics studies for a many-dimensional system is proposed. On the basis of a quantum-mechanical formulation of the problem a consequent derivation of the reaction rate equation along the line suggested by Eyring et al. [25] is developed which, however, is free of all restrictions of activated-complex theory. A more rigorous justification of the earlier proposed expression for the "characteristic temperature" for a system of n -dimensions is presented¹). Some general conclusions from the basic rate equation are also considered.

2. General Formulation

Assuming adiabatic separation between nuclear and electron motions (Born-Oppenheimer principle) we can represent the electronic energy of the system (i.e. the total energy except the kinetic energy of nuclei) as a function $V(x_1, x_2, \dots, x_n)$ of all nuclear coordinates $x_1, x_2, \dots, x_n$. A correction for the incomplete electron-nuclear separation may be introduced [9, 33] by replacing the electronic energy by an *effective* nuclear potential energy $V'(x_1, x_2, \dots, x_n)$ provided that the nuclear velocities are not very high so that the system remains in the same quasi-stationary electronic state.

The potential function $V(x_1, x_2, \dots, x_n)$ can be represented as a surface in a $(n + 1)$ -dimensional space. In the simplest case of a collinear triatomic reaction $AX + B = A + XB$ the potential energy $V(x_1, x_2)$ is a function of two independent internuclear distances $x_1 = r_{AX}$ and $x_2 = r_{XB}$; then, the potential surface lies in a three-dimensional space. The projection of this surface on the plane x_1, x_2 is represented by

¹) Some evidence for the validity of this expression for a two-dimensional potential-energy barrier is given in Ref. [9] and [24].

drawing several lines of constant energy (Fig. 1). Typically there are two "valley" R and P corresponding to reactants and products, respectively, and a "col" between them on the potential-energy surface [1–3]. We assume that this surface does not change qualitatively when replacing the electronic energy $V(x_1, \dots, x_n)$ by the effective potential energy $V'(x_1, \dots, x_n)$ accounting for the incomplete separation of nuclear and electron motions [9, 33]. This assumption is reasonable when the correction introduced in this way is not too large. When considering reactions in solution all coordinates describing the nuclear motions of both reacting particles and surrounding interacting solvent molecules have to be taken into account.

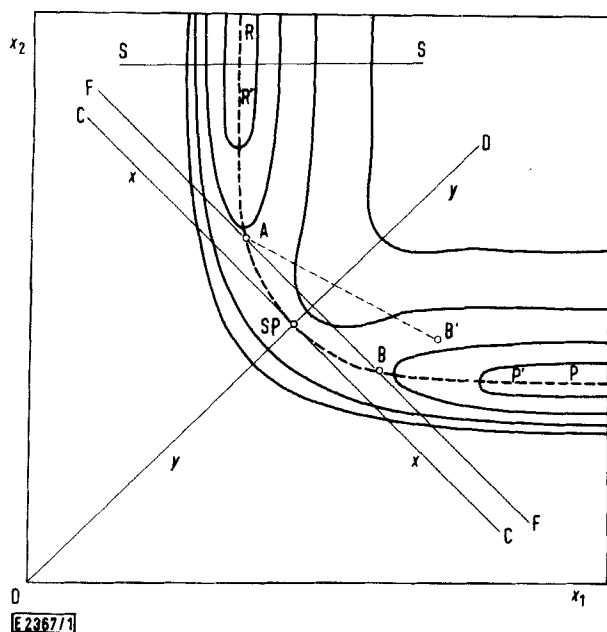


Fig. 1

Potential-energy map for the colinear triatomic reaction $AX + B \rightarrow A + XB$; $x_1 = r_{AX}$ and $x_2 = r_{XB}$, internuclear distances; R, reactants region; P, products region; SP, saddle-point; dotted line, classical reaction path; AB and AB', possible paths for nuclear tunneling

The "reaction path", i.e. the line of lowest energy goes from the reactants region (R) to products region (P) through the saddle-point SP. This line is in general a curved one. The reaction can be described classically by the translational motion of a representative point, with an effective mass μ^* , along a line near the reaction path called "reaction coordinate"²⁾. The energy profile along this line represents a potential curve with a maximum which is normally just at the saddle-point SP (Fig. 2). In gas-phase reactions this curve (Fig. 2a) becomes horizontal on both sides of the barrier in the regions R and P corresponding to free motions of reactants and products molecules, respectively. In solids and liquid solutions (Fig. 2b) the barrier along the reaction coordinate is situated

²⁾ The deviation of the classical "reaction coordinate" from the "reaction path" is a result of the centrifugal effect due to the curvature of the latter [26, 27]; this deviation depends also on the isotopic composition of the "activated-complex" [34].

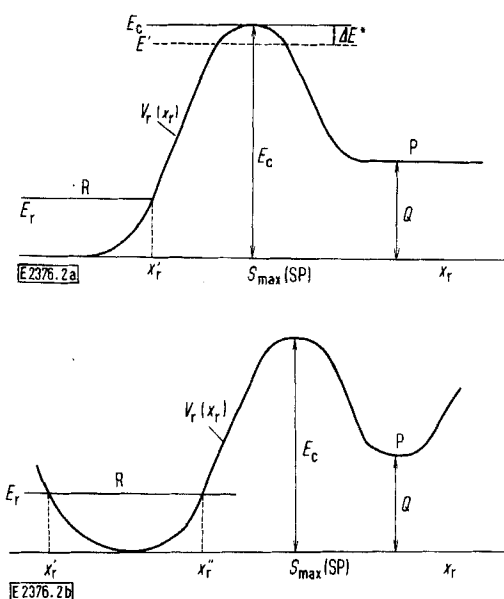


Fig. 2

Energy profile $V_r(x_r)$ along the reaction coordinate x_r : a) for gas-phase reactions, b) for reactions in solids and solutions; R, reactants region; P, products region; E_c , classical activation energy; Q , reaction heat; $E' = E_c - \Delta E^*$ (Equation (44)); x'_r, x''_r , turning points

between two potential wells in which vibrations occur in the initial and final states of the system, respectively.

From the point of view of classical mechanics the motion along the reaction coordinate leads from reactants to products when the system passes through the point of maximum potential energy ($S_{\max}$) which determines the "transition state". This point normally coincides with the position of the saddle-point³⁾. However, from the standpoint of quantum mechanics, in general, a departure from the classical reaction coordinate (say along the line AB or AB' in Fig. 1) is to be expected as a consequence of nuclear tunneling in the energy range below the top of the barrier corresponding to the point $S_{\max}$ (the saddle-point SP).

We describe the reacting system by a wave function $\psi(x_1, x_2, \dots, x_n)$ which is a solution of the Schrödinger equation

$$H\psi = E\psi \quad (1)$$

where H is the Hamiltonian including the potential energy $V(x_1, x_2, \dots, x_n)$ and E is the total energy of the system.

We now assume that in the initial and final states of the system the translational motion along the reaction coordinate can be separated from all other motions of reactants and products, respectively. This is always possible for all gas-phase reactions like the colinear triatomic reaction $AX + B = A + XB$. Thus, on Fig. 1 the linear portions RR' and PP' of the reaction coordinate⁴⁾ $\bar{x}_r$ represent the relative one-dimensional motions of reactants molecules (AX and B) and

³⁾ Sometimes the point of maximum potential energy along the reaction coordinate, corresponding to the "classical" transition state, may be different from the position of the saddle-point. In particular this point (and the saddle-point too) may lie very near to the products region (P).

⁴⁾ In these regions the "reaction coordinate" coincides with the "reaction path", i.e. the line of minimum energy.

products molecules (X and AB), respectively. In these regions the wave function can be expressed by the product

$$\psi(x_r, x_i) = \psi_r(x_r) \varphi(x_i) \quad (2)$$

where ψ_r describes the motion along the reaction coordinate x_r and φ describes all remaining motions, x_i denoting a corresponding set of $n-1$ coordinates (on Fig. 1 we have $x_r = x_2$ in the initial state and $x_r = x_1$ in the final state; $x_i = x_1$ or $x_i = x_2$ represents only one coordinate of vibration of reacting molecule AX or product molecule BX). For ψ_r one may use wave functions of the form

$$\psi_r(x_r) = \frac{C}{\sqrt{p_r}} e^{\pm \frac{i}{\hbar} \int p_r dx_r} \quad (3)$$

where p_r is the component of momentum along the reaction coordinate x_r . This is the form of the quasiclassical (WBK) approximation which holds under the condition [35]

$$\frac{\hbar \mu_r F_r}{p_r^3} \ll 1$$

in which μ_r is the reduced mass of the system and $F_r = -\partial V / \partial x_r$ is the force acting in x_r direction. This condition is fulfilled far enough from the saddle-point, where the potential V varies slowly along the reaction coordinate (Fig. 2a) if the momentum p_r is not very small (as is the case near the turning point x'_r at which $p_r = 0$). Very far from the barrier region, $F_r = 0$ and $p_r = \text{const}$ so that from (3) we obtain

$$\psi_r(x_r) = C' e^{\pm \frac{i}{\hbar} p_r x_r},$$

i.e. a plane wave which describes exactly the free (relative) translational motion of reacting particles (AX and B) or products molecules (XB + A).

For reactions in solids or in solution we may use again the WBK wave functions (3) in the regions of both potential wells (Fig. 2b), which refer to reactants and products, in order to represent the motion along the reaction coordinate (far enough from the turning points x'_r and x''_r); however, appropriate exact wave functions can also be used. In any case, the product solution (2) is to be considered either as an exact one or at least as a very good approximation for all chemical reactions.

Introduction of (2) into (1) gives two separate wave equations. The first one describes the motion along the reaction coordinate x_r :

$$H_r \psi_r = E_r \psi_r \quad (4)$$

with

$$H_r = -\frac{\hbar^2}{2\mu_r} \frac{\partial^2}{\partial x_r^2} + V_r(x_r), \quad E_r = \frac{p_r^2}{2\mu_r} + V_r(x_r),$$

where $V_r(x_r)$ is the potential along x_r and E_r the corresponding part of the total energy. The second equation refers to all remaining motions of reactants (or products):

$$H_n \varphi_n = E_n \varphi_n \quad (5)$$

where

$$H_n = -\frac{\hbar^2}{2} \sum_i \frac{1}{\mu_i} \frac{\partial^2}{\partial x_i^2} + V'(x_i), \quad (5a)$$

$$V'(x_i) = V(x_r, x_i) - V_r(x_r), \quad (5b)$$

$$E_n = E - E_r. \quad (5c)$$

Here n denotes a set of quantum numbers for reactants (or products), μ_i is the reduced mass corresponding to x_i , $V(x_r, x_i) \equiv V(x_1, x_2, \dots, x_n)$ is the full potential energy and E the total energy in (1); E_n is independent of x_r . We note that a separation of variables in (5) is *not* necessary for the following derivations.

We can now define the transition probability from reactants to products configuration. The systems moving in forward direction along the reaction coordinate in the reactants' region (R), being in an initial quantum state n , are partially reflected by the barrier returning back in another quantum state m and partially transmitted in the products' region (P) in a final quantum state n' . Thus, the wave functions ψ_i and ψ_f related to x_r -motion in the initial and final states are

$$\psi_i = \psi_r^{nm} = \psi_r^n + \psi_r^m, \quad (6a)$$

$$\psi_f = \psi_r^{n'}, \quad (6b)$$

where ψ_r^n and ψ_r^m represent an incident and a reflected wave in the reactants' region, respectively, and $\psi_r^{n'}$ is a transmitted wave in the products' region. We define the quantities

$$\kappa_{nn'} = \frac{I_n(\psi_r^{n'})}{I_n(\psi_r^n)}, \quad \rho_{nm} = \frac{I_m(\psi_r^m)}{I_n(\psi_r^n)}, \quad (7)$$

where

$$I = I_r = \frac{\hbar}{2\mu_r} \left(\psi_r^* \frac{d\psi_r}{dx_r} - \psi_r \frac{d\psi_r^*}{dx_r} \right) \quad (8)$$

is the statistical current density along the reaction coordinate. In (7) $\kappa_{nn'}$ is called "transfer coefficient" (or transition probability) and ρ_{nm} is the "reflection coefficient".

If the system is initially in a given quantum state n it can be reflected in any other initial state m or transmitted in any final state n' consistent with the total energy of the system. The condition of current conservation requires that

$$I_n - \sum_m I_m = \sum_{n'} I_{n'}, \quad (9)$$

since the total current density in forward x_r -direction (from reactants to products) must be the same on both sides of the barrier, summations being made over all quantum states of reactants and products consistent with the same value of total energy. From (7) and (9) it follows that

$$\sum_{n'} \kappa_{nn'} + \sum_m \rho_{nm} = 1. \quad (10)$$

3. Derivation of the Rate Equation

Since the motion along the reaction coordinate x_r in the initial state (reactants' region) can be always described by the quasi-classical wave function (3) we may apply classical statistics for this motions to calculate the rate constant.

Suppose the system is initially in a given quantum state n with respect to the x_i -motions. Then, the number of systems of this state to be on in the small interval between x_r and $x_r + dx_r$ of the reaction coordinate and in the corresponding momentum range between p_r and $p_r + dp_r$ is

$$dN_r(p_r) = N \frac{g_n e^{-E/kT}}{Z} \frac{dx_r dp_r}{h} \quad (11)$$

where according to (5c)

$$E = E_n + E_r; \quad (12)$$

N is the total number of systems (per unit volume), g_n is the statistical weight factor for the state n , Z is the full partition function of reactants and $dx_r dp_r/h$ is the number of quantum states in the phase-space element $dx_r dp_r$. Thus,

$$P_r dp_r \equiv \frac{1}{N} \frac{dN_r}{dx_r} = \frac{g_n e^{-(E_n + E_r)/kT}}{Z} \frac{dp_r}{h} \quad (13)$$

is the probability to find the system, being in the quantum state n , in unit interval of x_r and in the momentum range dp_r . On the other hand,

$$P_s^n dS = \frac{|\varphi_n|^2 dS}{\int |\varphi_n|^2 dS} \quad (14)$$

will represent the probability of finding the system (being in the same quantum state n) in the element $dS = \prod_i dx_i (x_i \neq x_r)$ of a hyperplane $S (x_r = \text{const})$ in the n -dimensional configuration space, normal to the reaction coordinate⁵⁾.

Multiplication of expressions (13) and (14) gives

$$P_n dp_r dS = P_r^n P_s^n dp_r dS \quad (15)$$

which is the probability for a system in state n to be simultaneously in the volume element $dr = dS dx_r = dS (dx_r = 1)$ of configuration space and in the momentum range dp_r . Then, the number of systems in state n crossing the element dS of the hyperplane S (far from the saddle-point!) in unit time is obtained⁶⁾ by multiplying (15) by the velocity v_r in x_r -direction (towards the saddle-point). Integration over dS gives the number of all systems crossing the hyperplane S in quantum state n with momentum p_r and multiplication by the transfer probability $\kappa_{nn'}(p_r)$, defined by (7) and (8), yields the fraction of these systems which react passing in final state n' . Finally, integration over p_r and summation over n and n' leads to a general equation for the rate constant

$$v = \int \int \sum_{p_r} \sum_{n'} P_n \kappa_{nn'} v_r dS dp_r = \int \sum_{n'} \sum_n P_r^n \kappa_{nn'}(p_r) v_r dp_r \int_S P_s^n dS. \quad (16)$$

Since in the reactants region, where (2) applies, φ_n depends only on $x_i (i \neq x_r)$ the integration over S using (14) can be separated from that over p_r so that one obtains

$$\int_S P_s^n dS = 1. \quad (17)$$

Further, from (12) we get the relation

$$v_r = \frac{\partial E}{\partial p_r} = \frac{\partial E_r}{\partial p_r}, \quad (18)$$

since E_n in (5) is independent on x_r and p_r , which permits to replace the p_r -integration by integration over E_r . In this way (16) with (13), (17), and (18) gives the rate expression

$$v = \frac{kT}{h} \frac{Z^*}{Z} \int \sum_n \sum_{n'} \kappa_{nn'}(E_r) f(E_n, T) e^{-E_r/kT} \frac{dE_r}{kT} \quad (19)$$

⁵⁾ On Fig. 1, referring to the colinear reaction $AX + B$, the hyperplane S reduces to a straight line $SS(x_r = x_2 = \text{const})$ normal to the reaction coordinate, in the reactants region R ; there, this is the line along which the vibration of molecule AX occurs.

⁶⁾ If we set in Equation (13) $N = 1$, i.e. assuming a concentration of one system per unit volume.

with

$$f(E_n, T) = \frac{g_n e^{-E_n/kT}}{Z^*}, \quad Z^* = \sum_n g_n e^{-E_n/kT}. \quad (20)$$

Here $f(E_n, T)$ is the occupation probability of a quantum state n and Z^* is the corresponding partition function of reactants.

We introduce now a new energy variable

$$\varepsilon = E_r - E_c$$

where E_c is the maximum potential energy along the reaction coordinate (normally the height of the saddle-point) or the so-called "classical activation energy" (Fig. 2), i.e. ε is the translational energy for x_r -motion (far from the saddle-point!) measured relative to E_c (instead relative to the minimum potential energy of reactants). Thus, (19) can finally be transformed as

$$v = \kappa \frac{kT}{h} \frac{Z^*}{Z} e^{-E_c/kT} \quad (21)$$

where

$$\kappa = \int_{-E_c}^{\infty} k(\varepsilon) e^{-\varepsilon/kT} d\varepsilon / kT, \quad (22)$$

$$k(\varepsilon) = \sum_n \kappa_n(\varepsilon) f_n(E_n, T), \quad \kappa_n = \sum_{n'} \kappa_{nn'}(\varepsilon).$$

Here κ_n is the total transition probability from a given initial quantum state n to any final state n' and $k(\varepsilon)$ is the statistical mean value of κ_n over all initial quantum states n . Thus, κ is defined as an average of $k(\varepsilon)$ in respect to ε in the whole range of variation of this magnitude.

The factor κ accounts for all quantum effects including tunneling through the barrier, reflection at the barrier and non-adiabatic transition from the lower to the upper potential surface. A complete separation of these effects is difficult to do, though it is possible to make it to a certain extent in somewhat arbitrary way. Thus, if the tunneling is negligible we have to set $k(\varepsilon) = 0$ for $\varepsilon < 0$ and $k(\varepsilon) = k'(\varepsilon)$ for $\varepsilon > 0$ so that (22) turns into

$$\kappa \rightarrow \chi = \int_0^{\infty} k'(\varepsilon) e^{-\varepsilon/kT} d\varepsilon / kT \quad (22a)$$

where χ accounts now only for the reflection due to the curvature of the reaction path and for the nonadiabatic transitions from the lower to the upper potential surface. Both effects can be treated separately using suitable quantum-mechanical methods [25, 35]. In particular, if the reaction is nonadiabatic the "transmission coefficient" χ can be calculated in the frame of perturbation theory (by applying, for instance, the method of Landau-Zener [35]) also when the nuclear tunneling is not negligible. In assuming, in principle, the possibility of an independent determination of χ we can define the "tunneling correction" by the ratio

$$\kappa_t = \frac{\kappa}{\chi}. \quad (23)$$

Evidently $\chi \leq 1$ and $\kappa_t \geq 1$ (since $\kappa \geq 1$). Thus, the rate Equation (21) can be written as

$$v = \kappa_t \chi \frac{kT}{h} \frac{Z^*}{Z} e^{-E_c/kT}. \quad (24)$$

In the classical limit we set in (24) $\kappa_i = 1$. If, moreover, the motion along the reaction coordinate is sufficiently slow an adiabatic adjustment of the remaining motions during reaction will assure a full quantization and a quasi-equilibrium energy distribution in the saddle-point region, too [4, 5]. Then, one is allowed to refer the partition function Z^* to the transition state⁷⁾. Under this conditions (24) turns into the usual equation of activated-complex theory.

4. The Characteristic Temperature

At sufficiently high temperatures the reaction will follow the classical reaction coordinate; then, in (24) $\kappa_i = 1$. In order to determine the condition on which the classical approximation is justified we may proceed in the following way: Using (22) and (23) the tunneling factor can be represented by the sum

$$\kappa_i = \kappa'_i + \kappa''_i = \frac{1}{\chi} \int_{-E_c}^0 k(\varepsilon) e^{-\varepsilon/kT} \frac{d\varepsilon}{kT} + \frac{1}{\chi} \int_0^\infty k(\varepsilon) e^{-\varepsilon/kT} \frac{d\varepsilon}{kT} \quad (25)$$

where the two terms correspond to transitions below and above the peak of the potential barrier along the reaction coordinate (normally the saddle-point). Similarly, according to (24) the rate constant can be written as

$$v = v' + v'' = (\kappa'_i + \kappa''_i) \chi \frac{kT}{h} \frac{Z^*}{Z} e^{-E_0/kT}. \quad (26)$$

Then, the condition

$$\frac{v'}{v''} = \frac{\kappa'_i(T)}{\kappa''_i(T)} = 1 \quad (27)$$

permits to define a "characteristic temperature" T_k at which the number of systems tunneling through the barrier ($\varepsilon < 0$) is just equal to that of the systems passing over the barrier ($\varepsilon > 0$). It is obvious from (22) and (23) that T_k will depend on the properties of the potential barrier through $k(\varepsilon)$, i.e. $\kappa_{nn'}$.

An approximate expression for T_k can be derived within certain assumptions as follows. It is to be expected that when condition (27) is fulfilled the majority of the systems pass through the saddle-point region near the classical reaction coordinate. In a small area around the saddle-point the reaction coordinate can be separated from the other coordinates (describing small harmonic vibrations in the transition state⁸⁾). There, we may use again the product wave function (2) for the whole system and the WBK approximation (3) to describe the motion along this short rectilinear part of the reaction coordinate. Choosing the origin of the coordinate system to be at the saddle-point SP we denote the reaction coordinate by x and the other coordinates by y_i . Thus, the "transition state"

is defined [26] by a hypersurface $S(y_i)(x = 0)$ passing through the saddle-point ($x = 0, y_i = 0$). (For the colinear system A-X-B this surface reduces to the line of highest energy OD denoted by y on Fig. 1.)

We consider a set of lines parallel to x -axis near the saddle-point. In any cross-cut along these lines ($y = \text{const}$)⁹⁾ the energy profile will represent a one-dimensional potential barrier, $V_y(x)$. We denote by $V_0(y)$ the minimum of this barrier on the reactants side and by U_x the total energy for motion in the x -direction measured relative to this minimum (Fig. 3). To calculate κ_i, κ'_i and κ''_i a quasi-equilibrium Boltzmann energy distribution will be assumed.

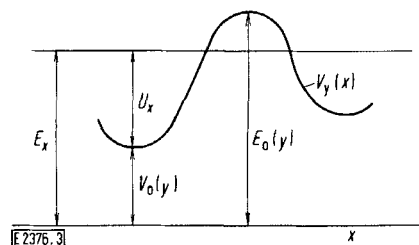


Fig. 3

Energy profile $V_y(x)$ in a cross-cut along a line $y = \text{const}$ parallel to x -axis (Fig. 1). $V_0(y)$, minimum potential-energy on reactants side; U_x , total energy for x -motion with respect to this minimum, E_x the same energy with respect to the minimum potential energy of reactants; $E_0(y)$ height of the one-dimensional barrier profile $V_y(x)$

If $W_y(U_x)$ is the probability for a transition in the energy level U_x through or over the barrier $V_y(x)$, then,

$$P'_y = \int W_y(U_x) e^{-U_x/kT} dU_x / kT \quad (28)$$

represents the total permeability of this barrier and

$$P_y = P'_y e^{-V_0(y)/kT} \quad (29)$$

is the probability for a system in the initial state to reach the energy level $V_0(y)$ and to pass along the line $y = \text{const}$ in the final state.

The total transfer probability from reactants to products is obtained by integration of (29) over the "transition state hypersurface" $S(y_i)$

$$P = \int P_y dS \quad (30)$$

where $dS = \prod_i dy_i$. Introducing the variable $E_x = V_0(y) + U_x$, from (28)–(30) one gets

$$P = \int dS \int_{V_0(y)}^\infty W_y(E_x) e^{-E_x/kT} dE_x / kT \quad (31)$$

where E_x is the energy measured relative to the minimum potential energy of reactants (Fig. 3).

Denoting the height of the one-dimensional barrier profile $V_y(x)$ along the line $y = \text{const}$ by $E_0(y)$ we find the classical transition probability by setting $W_y(E_x) = 0$ for $E_x < E_0(y)$ and $W_y(E_x) = 1$ for $E_x > E_0(y)$, hence

$$P_{cl} = \int dS \int_{E_0(y)}^\infty e^{-E_x/kT} dE_x / kT = \int e^{-E_0(y)/kT} dS. \quad (32)$$

⁹⁾ For simplicity we denote by y the set of coordinate y_i defining the transition state hypersurface.

⁷⁾ This implies separation of the reaction coordinate from the other coordinates according to (2), (4) and (5) at least in a small region around the saddle-point; then, E_n will be the energy in the transition state except for motion along the reaction coordinate.

⁸⁾ The reaction coordinate coincides with one of the normal coordinates of vibration in the transition state where it describes an unstable vibration with an imaginary frequency ν_i (it corresponds, for instance, to the asymmetric stretching vibration of the triatomic complex A–X–B). The deviation of this normal coordinate from the classical reaction path (i.e. the line of lowest energy) is normally very small [34].

Thus, the tunneling correction κ_t is given by the expression

$$\kappa_t = \frac{P}{P_{cl}} = \frac{\int F(y) dS}{\int e^{-E_0(y)/kT} dS} \quad (33)$$

where

$$F(y) = \int_{V_0(y)}^{\infty} W_y(E_x) e^{-E_x/kT} dE_x / kT.$$

$F(y)$ has a maximum at a certain point y_0 on the "transition state hypersurface" $S(y_i)$ (Fig. 4). The most probable line ($y_i = y_0$) will cross $S(y_i)$ very near to the saddle-point ($x = 0$, $y_i = 0$), when condition (27) is approximately valid, i.e.

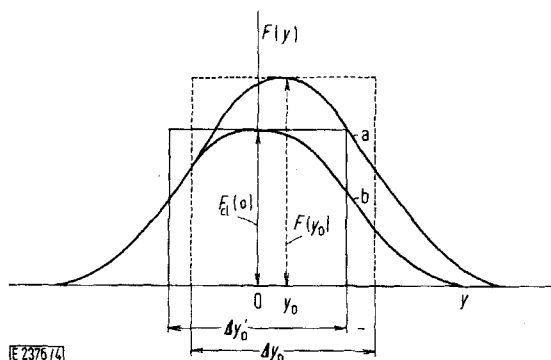


Fig. 4

The integrands in Equation (33) in one-dimensional representation ($dS = dy$); curve a describes $F(y)$ and curve b represents $F_{cl}(y) = e^{-E_0(y)/kT}$. The areas under the curves a and b equal $F(y_0)\Delta y_0$ and $F_{cl}(0)\Delta y'_0$, respectively ($\Delta y_0 \simeq \Delta y'_0$)

$v'/v'' \sim 1$; this is also the case when $v'/v'' < 1$ (i.e. the number of systems surmounting the barrier is greater than that of the tunneling systems). Then, $F(y)$ tends to zero rapidly with increasing distance from $y = 0^{(1)}$ (Fig. 4) so that we can write

$$\int F(y) dS = F(y_0) \Delta S_0 \quad (34)$$

where ΔS_0 is a small area around y_0 in which $F(y)$ is most significant. The maximum value of $\exp(-E_0(y)/kT)$ is at $y = 0$; there $E_0(0) = E_c$ is just the height of the saddle-point. Hence,

$$\int e^{-E_0(y)/kT} dS = e^{-E_c/kT} \Delta S'_0. \quad (35)$$

We may set $\Delta S'_0 \simeq \Delta S_0$ when $v'/v'' \lesssim 1$ (Fig. 4) so that from (33)–(35) we obtain

$$\kappa_t = e^{E_c/kT} \int_{V_0(y_0)}^{\infty} W_{y_0}(E_x) e^{-E_x/kT} \frac{dE_x}{kT} \quad (36)$$

as an approximate expression for the tunneling correction.

By decomposing the integral in (36) in two terms, as in (26), corresponding to systems tunneling and surmounting the barrier $V_{y_0}(x)$, we can write

$$\frac{\kappa'_t}{\kappa''_t} = \frac{\int_{V_0(y_0)}^{\infty} W_{y_0}(E_x) e^{-E_x/kT} dE_x}{\int_{E_0(y_0)}^{\infty} W_{y_0}(E_x) e^{-E_x/kT} dE_x} = 1 \quad (37)$$

¹⁰⁾ This situation is demonstrated using two-dimensional barrier models [40] (see also Ref. [24]).

which is like the condition (27). At this condition we may set $y_0 = 0$ and $E_0(0) = E_c$, since the most probable line ($y = y_0$) is near the saddle-point position and the barrier profile $V_y(x)$ varies slowly with y in the vicinity of this point. Then (37) will correspond fully to (27) defining the characteristic temperature T_k .

The one-dimensional tunneling probability $W_0(E_x)$ can be evaluated in a quasi-classic approximation using the expression [12, 14, 18, 28, 29, 37]

$$W_0(E_x) = \frac{1}{1 + e^{H(E_x)}}, \quad H(E_x) = \frac{4\pi}{h} \sqrt{2\mu^*} \int_{x_1}^{x_2} \sqrt{E_x - V_0(x)} dx, \quad (38)$$

where μ^* is the effective mass for motion in x -direction, x_1 and x_2 are the classical turning points (at which $E_x = V(x)$). Equations (36) and (37) are valid when the systems pass predominantly in an energy range near the barrier top in which $V_0(x)$ may be well approximated by a parabolic function (Fig. 2)

$$V_0(x) = E_c - \frac{f_x^*}{2} x^2 \quad (39)$$

where the force constant $f_x^* = -(\partial^2 V / \partial x^2)_{x=0}$ is determined by the curvature of the barrier profile just at the saddle-point. Integration in (38) using (39) yields [18]

$$H(E_x) = 2 \frac{E_c - E_x}{k T_k} \quad (40)$$

where

$$T_k = \frac{h}{2\pi^2 k} \sqrt{\frac{f_x^*}{\mu^*}} = \frac{h\nu^*}{\pi k}, \quad \nu^* = \frac{1}{2\pi} \sqrt{\frac{f_x^*}{\mu^*}}, \quad (41)$$

ν^* being the frequency of a harmonic vibration with the force constant f_x^* .

Using (38)–(41) to evaluate the two integrals in (37) it can be shown [16] that they are equal just at a temperature $T = T_k$ given by (41). Thus, this is the "characteristic temperature", originally defined by the condition (27), at which the number of systems tunneling through the barrier ($E_x < E_c$) equals the number of systems surmounting it ($E_x > E_c$). The magnitude

$$\Delta E^* = k T_k = \frac{h\nu^*}{\pi} \quad (42)$$

represents a characteristic energy for a given reacting system. The physical meaning of this magnitude will be clarified by the following consideration.

By means of (38)–(41) we find that the integrand in (36) has a maximum at the energy value

$$E_m = E_c - \frac{k T_k}{2} \ln \frac{\frac{1}{2}(T_k/T)}{1 - \frac{1}{2}(T_k/T)}. \quad (43)$$

This equation is valid for $T > T_k/2$. We see that when $T = T_k$, $E_m = E_c$, i.e. the energy distribution of the transmitted systems is symmetrical in respect to the barrier peak (along the reaction coordinate). Further, defining by means of (42) the energy value [18b]

$$E' = E_c - \Delta E^* = E_c - k T_k \quad (44)$$

we find that $E_m = E'$ when $T = [(1 + e^2)/2e^2] T_k = 0.567 T_k$. Assuming that in the energy range $E_x > E'$ (Fig. 2a) the

parabolic function (39) is still a good approximation for the barrier profile $V_0(x)$ we obtain, integrating in (36), the expression¹¹⁾

$$\kappa_t = \frac{(\pi/2)(T_k/T)}{\sin[(\pi/2)(T_k/T)]}. \quad (45)$$

This expression is valid for $T > T_k/2$ but the condition $T > 0.567 T_k$ implies that it is expected to give satisfactory results when most of the systems pass through and over the barrier in the energy range $E > E'$ where E' is defined by (44). It has been shown, indeed, using both one-dimensional and two-dimensional barrier models [9, 24, 30], that the tunneling correction can be calculated sufficiently accurate by means of (45) when $T > 2/3 T_k \approx 0.7 T_k$.

We find the following typical values for κ_t in the region of validity of (45) ($T > T_k/2$) (Table 1)

Table 1		
$T \geq 2 T_k$	$T = T_k$	$T \leq 0.567 T_k$
$\kappa_t \leq 1.1$	$\kappa_t = 1.57$	$\kappa_t \geq 6.9$

These results permit to divide conventionally the whole temperature range in three intervalls corresponding to the conditions of large tunneling, moderate tunneling and negligible tunneling as shown in Table 2.

Table 2		
large tunneling region: $T < T_k/2$	moderate tunneling region: $T_k/2 < T < 2 T_k$	classical region: $T > 2 T_k$
$\kappa_t > 7$	$\kappa_t \approx 1$ to 7	$\kappa_t = 1$

We see that the one-dimensional treatment of the reaction coordinate is justified in the temperature region of moderate tunneling, where expression (45) for κ_t may be used, as well as in the region of negligible tunneling, where classical mechanics applies. However, in the region of large tunneling, in general, a many-dimensional treatment is necessary to compute sufficiently accurately the quantum corrections to the classical reaction rate [20–24]. Nevertheless, many qualitative conclusions based on the one-dimensional approximation are certainly generally valid in chemical kinetics.

5. Some General Consequences

From the fundamental rate Equation (24) several general conclusions can be drawn without introducing any restrictions. By a standard transformation we can write this equation as

$$v = \kappa_t K e^{-E_a/kT} \quad (46)$$

¹¹⁾ Expressions of the type of (45) have been derived on the basis of a parabolic potential function by several authors [12, 14b, 37, 38]. The inclusion of the characteristic temperature T_k provides a more convenient form of (45) for determining the extent of tunneling [18b] as shown also in the text.

with

$$K = \chi \frac{kT}{h} \frac{Z'^{\ddagger}}{Z'}, \quad E_a = E_c + (\varepsilon_0^{\ddagger} - \varepsilon_0)$$

where the new partition functions Z' and $Z'^{\ddagger}$ include the vibrational factors $Z'_v = \prod_i (1 - e^{-h\nu_i/kT})$ corresponding to the initial and transition state, respectively, E_a is the "true" classical activation energy, ε_0 is the sum of zero-point energies of reactants and $\varepsilon_0^{\ddagger}$ is some effective zero-point energy of the transition state¹²⁾ (with a corresponding effective frequency $\nu^{\ddagger} = \varepsilon_0^{\ddagger}/h$ which enter $Z'^{\ddagger}$).

The experimentally determinable kinetic parameters in the Arrhenius equation $v = K' \exp(-E'_a/kT)$ defined by

$$E'_a \equiv -k \frac{\partial \ln v}{\partial (1/T)}, \quad K' \equiv v e^{E_a/kT}$$

give some effective values for the activation energy (E'_a) and preexponential factor (K'). From (46) we find

$$E'_a = E_a - k \frac{\partial \ln K}{\partial (1/T)} - k \frac{\partial \ln \kappa_t}{\partial (1/T)}, \quad (47a)$$

$$K' = \kappa_t K e^{(E'_a - E_a)/kT}. \quad (47b)$$

Since K is in general slightly dependent on temperature the second term in (47a) can be usually neglected. Thus, $E'_a - E_a$ and K'/K are essentially determined by the tunneling correction κ_t and its variation with temperature.

With increasing temperature κ_t decreases, hence $E'_a < E_a$, i.e. the experimental activation energy is always lower than the classical one when tunneling is not negligible ($\kappa_t > 1$). Since K'/K depends on both $\kappa_t > 1$ and $E'_a - E_a < 1$, from the general form of (47b) it cannot be decided whether $K' < K$ or $K' > K$. This is possible, however, in the region of moderate tunneling ($T > T_k/2$) using expression (45) for κ_t which gives¹³⁾

$$E'_a = E_a - kT \left(1 - \frac{\pi\gamma}{2} \cot \frac{\pi\gamma}{2} \right), \quad (48)$$

$$K' = K \frac{\pi\gamma/2}{\sin(\pi\gamma/2)} \exp \left[- \left(1 - \frac{\pi\gamma}{2} \cot \frac{\pi\gamma}{2} \right) \right],$$

where $\gamma = T_k/T$. It results that not only $E'_a < E_a$ but also $K' < K$, and the difference between the absolute values of E'_a and E_a as well as between K' and K increases with increasing γ (i.e. the tunneling degree). These equations predict a deviation from the linear Arrhenius plot $\ln v = f(1/T)$. All these conclusions are shown to be valid in the region of large tunneling ($T < T_k/2$) using both one-dimensional [15b] and two-dimensional [40] barrier models.

¹²⁾ The existence of some zero-point energy in the saddle-point region is a consequence of the uncertainty principle [39] $\Delta p_i \Delta x_i \sim h$ ($x_i \neq x_i$, $p_i \neq p_i$) since the momentum p_i cannot be zero there ($\Delta p_i \neq 0$ because $\Delta x_i \neq \infty$). This is true also when, because of the very short lifetime $\Delta\tau$ of the transition state, the uncertainty of the energy determination $\Delta\varepsilon \sim h/\Delta\tau$ is too large and may be comparable with the value of the zero-point energy corresponding to a stationary quantum state in the transition region [9]. Therefore an effective value of this energy should be introduced.

¹³⁾ Equations (48) are identical with the expressions first derived by Bell [12] for a one-dimensional parabolic barrier.

From (46) a general equation for the kinetic isotopic factor is obtained

$$S \equiv \frac{v_H}{v_D} = \frac{\kappa_H}{\kappa_D} \frac{K_H}{K_D} e^{-(\epsilon_{a,H} - \epsilon_{a,D})/kT} \quad (49)$$

where the indices refer to two isotopes (for instance H and D). Since $\kappa_H \kappa_D$ is usually near to unity, S depends essentially on the ratio of tunneling corrections and on the difference of zero-point energies for the two isotopes. Assuming that the reaction occurs in the region of moderate tunneling, we can estimate on the basis of experimental data for E'_a, K' , and S the tunneling correction and compute the characteristic temperature T_k using (45), (48), and (49) without knowledge of the potential-energy surface [9, 15b, 17].

Another general consequence concerns the relation between activation energy and reaction heat Q . We define the quantities

$$\beta' \equiv \frac{\partial E'_a}{\partial Q}, \quad \alpha' \equiv -\frac{\partial(E'_a - Q)}{\partial Q} \quad (50a)$$

as experimental "transfer coefficients" and

$$\beta \equiv \frac{\partial E_a}{\partial Q}, \quad \alpha \equiv -\frac{\partial(E_a - Q)}{\partial Q} \quad (50b)$$

as "classical" transfer coefficients through the experimental activation energies (E'_a and $E'_a - Q$) and classical activation energies (E_a and $E_a - Q$) for forward (endothermic) and reverse (exothermic) reaction, respectively. Then, from (47a) we obtain the relations

$$\beta' = \beta + kT^2 \frac{\partial^2 \ln \kappa_t}{\partial T \partial Q}, \quad (51)$$

$$\alpha' = \alpha - kT^2 \frac{\partial^2 \ln \kappa_t}{\partial T \partial Q}.$$

In the region of moderate tunneling ($T > T_k/2$), where κ_t according to (45) is independent on Q , we have $\beta' = \beta$ and $\alpha' = \alpha$. However, in the region of large tunneling ($T < T_k/2$) the difference between β' and β (α' and α) may be considerable.

Further, from (46), (50), and (51) using the equilibrium constant expression

$$K_c = C e^{-Q/kT}$$

we derive the Brönstedt equations

$$v_1 = C_1 K_c^{\bar{\beta}}, \quad v_2 = C_2 K_c^{-\bar{\alpha}} \quad (52)$$

where v_1 and v_2 are the rate constants for endothermic and exothermic reaction, respectively. Here, $\bar{\beta}$ is generally different from both β' and β , and $\bar{\alpha}$ is different from α' and α . Only when the tunneling correction κ_t is independent of the reaction heat Q , i.e. when the tunneling is moderate ($T > T_k/2$), we find that $\bar{\beta} = \beta' = \beta$ (and $\bar{\alpha} = \alpha' = \alpha$). If the reaction occurs in the region of large tunneling ($T < T_k/2$) the differences between $\bar{\beta}$, β' , and β may be rather large. These differences may be significant also in the transition range between moderate and large tunneling ($T \sim T_k/2$) where the formula (45) for κ_t is not accurate and κ_t is generally slightly dependent on Q [14b, 18b]. This situation has been demonstrated using one-dimensional models for the barrier profile along the reaction coordinate [9].

Since the transfer coefficients α' and β' can be determined experimentally using (50a) independently from the Brönstedt coefficients $\bar{\alpha}$ and $\bar{\beta}$ in (52), a possible difference between α' and $\bar{\alpha}$ (β' and $\bar{\beta}$) will be an indication for a considerable role of nuclear tunneling. Thus, an experimental criterion for determining the tunneling degree is found. In particular it permits to conclude that the reaction occurs in the region of moderate tunneling (at $T > \frac{2}{3} T_k$) if $\alpha' = \bar{\alpha}$ ($\beta' = \bar{\beta}$): then, expression (45) for κ_t may be used as a good approximation.

6. Conclusions

The first main result in this paper (Sec. 3) is the derivation of the rate Equation (21) which is valid without any restrictions for a n -dimensional system. It represents a generalization of an earlier treatment [13, 14] in which the reaction is described in one dimension. Expression (21) has the usual form of the basic equation of activated-complex theory, however, it differs essentially from the latter in two points: First, the partition function $Z^\ddagger$ refers to the reactants and not to the transition state. Second, it contains a factor κ which accounts quite generally for all quantum effects. The rate equations derived in other closely related works have not a such general validity because of restrictions involved either in the basic assumptions or introduced in the course of the derivation. Thus, Eyring et al. [25], although starting from a quite general formulation, finally arrive to a rate expression in which the tunneling correction κ_t is completely omitted¹⁴). Marcus [26, 27], on the other hand, taking into account the nuclear tunneling in an extended but though limited region around the saddle-point, has obtained a rate equation on the basis of a vibrationally adiabatic approximation equivalent to the equilibrium postulate of activated-complex theory.

The rate expression (21) has been derived by separating the reaction coordinate in the initial state [25] and not, as usual, in the transition state [1, 4] of the system. Introducing of curvilinear coordinates [26, 27] is therefore not necessary for this purpose¹⁵). In this way it was shown in the present paper, how a quite general rate equation of the familiar form can be derived.

The possibility of a consequent derivation of the basic equation of chemical kinetics without introducing any restrictive conditions is certainly more satisfactory from logical point of view than all similar earlier derivations [1, 4, 5, 13–15, 25, 26, 27, 34]. Apart from this advantage, in principle, the present treatment leads in a natural way to an expression for the quantum correction κ , Equation (22), which can be used to calculate *directly* this quantity¹⁶). This way is essentially different

¹⁴) Eyring et al. have really written a general expression (Ref. [25], Equation (16.15)), however, they assert that it is impossible to simplify it without introducing some assumptions leading to the classical approximation. Thus an equation of the form (24) with $\kappa_t = 1$ is obtained.

¹⁵) The curvilinear coordinates may be very useful for calculation of reaction rates, including tunneling, within the conditions of validity of the rate equations derived by Marcus [26, 27].

¹⁶) For this purpose use can be made of the possibilities of electronic computers for numerical solving the wave equation [23] in order to evaluate the transfer coefficients $\kappa_{nn'}$ from Equations (7) and (8) (Ref. [41]).

from the approach based on the usual definition, $\kappa = v/v_{cl}$, which implies an independent evaluation of *two* quantities (v and v_{cl}), i.e. both a quantum-mechanical and a classical calculation of the rate constant; thereby, some assumptions are inevitably involved, for instance, the equilibrium hypothesis of activated-complex theory in evaluating v_{cl} as done by Mortensen [23]. The direct application of Equation (22) allows to avoid such restrictions when calculating the quantum correction κ [41].

An other important result (Sec. 4) is the formula (41) for the "characteristic temperature" of a system of n -dimensions which is similar to an earlier found expression for the one-dimensional case [14, 16, 18]. For the purpose of the present more general derivation the usual separation of the reaction coordinate in the vicinity of the saddle-point is used as a good approximation. Thus, a detailed analysis shows well the conditions at which the n -dimensional tunneling problem reasonably enough reduces to a one-dimensional one. The characteristic temperature T_k proves to be a simple and very suitable criterion for determining the conditions within which a quantum-mechanical treatment of reaction rates is really necessary or, alternatively, a classical treatment is sufficiently accurate. In this way it is useful to introduce a conventional delimitation of the temperature regions in which the nuclear tunneling plays either a large or a moderate role, or may be completely neglected. It is shown that in the region of moderate tunneling ($T_k/2 < T < 2T_k$) the tunneling correction κ_i can be conveniently expressed in terms of T_k by the simple formula (45) which has been often used in a different form [12, 14, 34]; thus, the conditions of its validity for a n -dimensional system are well defined by the present analysis.

It is not unimportant to show (Sec. 5) that some well known conclusions of earlier more special investigations [11, 15] follow directly in a quite general manner from the basic rate expression (24), hence they are free to the same extent of any restrictive assumptions. This concerns the Equations (47) for the experimental activation energy and frequency factor, and expression (49) for the isotopic separation factor as well. A quite new general consequence is represented by the relation (51) between the "experimental" and "classical" transfer coefficients which determine the dependence of the corresponding activation energies on heat of reaction. It is important to stress not only that these quantities differ, in general, from one another, but also that they are both different from the coefficients in the Brönstedt relations (52) between rate constants and equilibrium constant. This difference provides a new criterion for determining the extent of nuclear tunneling on the basis of experimental studies.

Note added in proof: Quite recently Truhlar and Kuppermann [42] have shown that the exact quantum mechanical calculations for the rate constant of the reaction $H_2 + H = H + H_2$ do not agree with activated-complex theory at relatively low temperatures. This disagreement is explained both by the inadequacy of the equilibrium assumption for the real vibration in the transition state and the inaccuracy of the one-dimensional tunneling corrections commonly used. These conclusions support, in general, the basic ideas of the present paper. However, the treatment of Truhlar and Kuppermann,

like that of Mortensen [23], does not give directly the two-dimensional tunneling corrections as discussed in Sec. 6 of the present paper.

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Reaktion von O(³P)-Atomen mit Allen

Von P. Herbrechtsmeier und H. G. Wagner

Aus dem Institut für Physikalische Chemie der Universität Göttingen

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In einem isothermen Strömungssystem wurde die Reaktion von O(³P)-Atomen mit Allen bei 5 Torr und Temperaturen von 275 bis 375 K im Trägergas Helium untersucht. Als Hauptprodukte treten C₂H₄ und CO auf, die sehr wahrscheinlich direkt gebildet werden. Für die Geschwindigkeitskonstante der Reaktion $O + C_3H_4 \rightarrow C_2H_4 + CO$ ergab sich

$$k \approx 7,8 \cdot 10^{12} \exp\left(-\frac{E}{RT}\right) \frac{\text{cm}^3}{\text{mol sec}} \text{ mit } E = 1600 \text{ cal/mol}$$

The reaction of O(³P) atoms with allene has been investigated in an isothermal flow system at 5 torr and temperatures between 275 and 375 K with He as carrier gas. Main products are C₂H₄ and CO. For the reaction $O + C_3H_4 \rightarrow C_2H_4 + CO$ a rate constant of

$$k \approx 7,8 \cdot 10^{12} \exp\left(-\frac{E}{RT}\right) \frac{\text{cm}^3}{\text{mol sec}} \text{ with } E = 1600 \text{ cal/mol}$$

has been obtained.

Einleitung

Die Reaktionen von O-Atomen mit Olefinen verlaufen unter Bildung von Additionscomplexen, die bei hinreichend niedrigen Drücken isomerisieren (z. B. durch Wanderung eines H-Atoms), und zerfallen [1]. Auch bei der Reaktion von O-Atomen mit Alkinen kann ein H-Atom wandern, z. B. $C_2H_2 + O \rightarrow CH_2 + CO$ [2]. Diese Reaktionen, besonders wenn sie mit größeren Kohlenwasserstoffmolekülen ablaufen, können über verschiedene Reaktionskanäle ablaufen. Im Rahmen einer Untersuchung der Reaktionen von O-Atomen mit ungesättigten C₃-Kohlenwasserstoffen wurde auch die Reaktion von O-Atomen mit Allen gemessen. Über einige Beobachtungen an dieser Reaktion soll hier berichtet werden.

Versuchsanordnung

Die Apparatur besteht aus einem thermostatierten Strömungssystem aus Pyrexglas [3] (Länge 22 cm, Innendurchmesser 2,2 cm), das über ein neu konstruiertes Molekularstrahlabsaugsystem an ein Bendix-Flugzeitmassenspektrometer angeschlossen ist [4].

O-Atome wurden in einem Seitenarm des Strömungssystems in einer elektrodenlosen Mikrowellenentladung aus O₂ oder NO erzeugt (<1% O-Atome in He). Durch eine verschiebbare Sonde wird Helium mit <10% Allen zugegeben. Der Druck betrug 5 Torr, die Strömungsgeschwindigkeit 19 m/sec. Zur Messung des O-Atomabfalls wurde eine Elektronenenergie von 25 eV, für die stabilen Produkte 70 eV gewählt.

Meßergebnisse

Reaktionsprodukte: Die größten Zunahmen während der Reaktion traten auf den Massenlinien 26, 27 und 28 auf. Weiter wurden geringe Zunahmen auf den Massen 30 und 29 (gemessen mit O aus O₂), 42 und 43 sowie 56, dem Additionsprodukt von O mit Allen, und 55 beobachtet.

Eine genaue Zuordnung der Massenlinien mit Hilfe von Eichspektren erwies sich als schwierig. Daher wurden die stabilen Produkte der Reaktion gaschromatographisch analysiert. Der benutzte Säulentyp (62, S 9,37 Temp. 50°C) eignete sich zum Nachweis von CH₄, C₂H₂, C₂H₄, C₂H₆, C₃H₈, Cyclopropan, C₃H₆, Allen, Buten-1 und Isobuten. Bei hohem Allenüberschuß [Allen : O ≈ 15] konnte von diesen Kohlenwasserstoffen nur C₂H₄ nachgewiesen werden. Die Nachweisgrenze für C₂H₂ betrug 7% der C₂H₄-Konzentration. Zur Prüfung auf CO wurden die unter gleichen Bedingungen erhaltenen Proben mit einem hoch auflösenden Massenspektrometer analysiert. Es ergab sich eine Ausbeute von 97 Molprozent CO und 87 Molprozent C₂H₄ bezogen auf einen Umsatz von 100 Molprozent Allen.

Somit sind CO und Äthylen die Hauptprodukte der Reaktion. Werden nur diese beiden Produkte für die Massenbilanz der C-Atome berücksichtigt, dann fehlen ca. 10% der Kohlenstoffatome, ein Wert, der etwas außerhalb der Fehlergrenzen liegt. Sucht man diesen Restbetrag durch Zuordnung der übrigen auftretenden Massenlinien auszugleichen, dann ist vor allem auf die Massen 29 und 30 hinzuweisen. Das Verhältnis dieser Massenlinien entspricht dem Bruchstückverhältnis von Formaldehyd, ein Hinweis auf das Auftreten einer HCHO-Gruppierung als Molekül oder als Molekülfragment.

Die Zahl der pro Allenmolekül in der Reaktion verbrauchten O-Atome nimmt mit zunehmendem Allenüberschuß zu und nähert sich bei hohen Allenüberschüssen dem Wert eins.

Messung der Geschwindigkeitskonstanten

Zur Bestimmung der Geschwindigkeitskonstanten wurden O(³P)-Atome aus NO erzeugt, so daß in einer von molekularem Sauerstoff relativ freien Atmosphäre (10% O₂ bezogen auf NO) gemessen werden konnte. Die Abnahme der O-Atome wurde bei einem C₃H₄/O-Verhältnis von 15 gemessen. Dort ist bei Zimmertemperatur $\Delta O/\Delta C_3H_4 \leq 1,1$. Bei diesem Überschuß von Allen kann der Allenverbrauch im Rahmen der Genauigkeit der hier verwendeten Versuchsanordnung vernachlässigt werden.